Random Load Pattern Recognition of Test Road Based on a Laser Direct Writing Carbon-Based Strain Sensor and a Deep Neural Network

Kuan Ju, Shuo Weng, Chun Li, Lihui Zhao, Bo Li, Yue Hu, Yang Gao, and Fuzhen Xuan

***Abstract*— The practical application of hydrogen-based fuel cell vehicles (HFCVs) requires the design of advanced stress monitoring system on the hydrogen storage devices, relying on accurate feedback and recognition from the vehicles load conditions. Herein, we proposed an intelligent monitoring method for continuous monitoring of dynamic load parameters and load recognition by combining carbon-based strain sensor, data acquisition and wireless transmission embedded unit, and machine learning (ML). The carbon-based strain sensor, prepared by laser direct writing (LDW) on polyimide (PI) film, exhibited a sensitivity of 6.08 and a frequency response of 1 Hz for effectively measuring the strain status under varying road conditions. A miniaturized and highly integrated data acquisition and wireless transmission unit was adopted to implement edge computing and wireless communication. Wavelet packet decomposition (WPD) and Hilbert–Huang transform (HHT) were fused to extract intrinsic features in random load spectrum, and the distance-based feature evaluation method was used to evaluate sensitive features to optimize the structure of the deep neural network (DNN). The DNN model could effectively recognize six types of test road conditions, with a test accuracy over 90%. This research effectively advances the design of the loading condition monitoring system of the HFCVs hydrogen storage device.**

***Index Terms*— Deep neural network (DNN), edge computing, load condition recognition, signal processing, strain sensor.**

I. Introduction

FROM the perspective of environmental sustainability and the global energy shortages, hydrogen-based fuel cell vehicles (HFCVs) have shown its promising potential to be an important part of future vehicles [1]. Although hydrogen has the characteristics of clean consumption, high-efficiency, and sufficient reserves, safe storage and transportation of hydrogen is still a huge obstacle to the development of HFCV industry. High-pressure hydrogen storage vessels (HSVs) may suffer unforeseen failure during long-term use, resulting in severe accidents, such as fatigue failure, gas leakage, filament wound breakage, hydrogen embrittlement, and so on [2], [3]. Particularly, fatigue failure caused by random loads under different road conditions is one of the most important causes of various failures. To prevent such accidents, an accurate, real-time, and long-term wireless monitoring of the load status of vehicle-mounted HSV is indispensable.

Traditional diagnosis methods based on manual periodic inspections rely on human senses and experience to perceive the vibrations of the equipment and subjectively evaluate its operating status, which cannot meet the requirements for real-time monitoring. Sensor-based structural health monitoring (SHM) technology can continuously and reliably acquire structural response and load signals and has been widely employed in actual infrastructure [4]. The mainstream sensors used for SHM include optical fibers, acoustic emission, strain gauges, and so on. Optical fiber-based SHM techniques have been widely utilized in civil engineering, aerospace, and other industrial fields [5], [6]. It has the advantages of high precision and good stability, whereas the optical demodulation equipment is expensive and the demodulation processing is complicated [7]. Acoustic emission sensing has powerful potential in structural damage monitoring, yet the influence of material microstructure and environmental noise on measurement accuracy limits its industrial application [8]. In contrast, strain gauges are more cost-effective and reliable, making them popularly used in SHM of automobiles, wind turbine blades, and rotors [9], [10].

Commercial strain gauges measure deformation through resistance changes caused by geometry changes. Thus, the sensitivity of the device depends on the Poisson's ratio, resulting in the gauge factors typically less than 2 [11]. To improve the sensitivity of strain sensors, nanomaterials, such as carbon nanotubes, silver nanowires, graphene, and so on [12], [13], [14], have been employed to develop strain sensors. Deformation-induced changes in the conductive network of the nanomaterial enable strain sensors with high sensitivity (up to 100) and rapid response (∼100 ms) [14], [15]. Despite these achievements, scalable and cost-effective preparation techniques are also

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required to realize the application of these nanomaterial-based sensors in industrial scenarios. Methods, such as 3-D printing [16], inkjet printing [17], and screen printing [18], have been proposed for the fabrication of the nanomaterial-based strain sensors. However, preparing inks with proper conductivity and viscosity for these methods is time-consuming and labor-intensive. In this study, a laser direct writing (LDW) [19] technology was used to prepare the flexible carbon-based sensor in a maskless, highly efficient, and large-scale manner.

Additionally, research on nanomaterial-based sensors has primarily focused on improving sensor performance, and there is a lack of discussion on the subsequent data acquisition, transmission, and processing. Strain gauge-based wireless sensor networks (WSNs) have been actively explored as an alternative to traditional wiring maintenance in recent years [20]. Many WSN systems utilize centralized big data processing methods based on cloud computing, while the transmission and processing of massive data from edge devices may lead to system response delay, power consumption increase, and data privacy leakage [21]. Accordingly, this research presents an edge wireless lightweight measurement system, which realizes the acquisition, preprocessing, and wireless transmission of load signals, thereby reducing response delays, simplifying wiring layout, and improving data security.

Due to the complexity of the HFCV structure, the distribution and probability statistics of damage are not obvious. In addition, the dynamic time-varying and coupling characteristics of road random load spectra and invalid noise signals all pose challenges to traditional sensing and monitoring technologies. Traditional time- and frequency-domain signal analysis methods, which are usually based on the assumption of a stationarity signal, cannot simultaneously consider the global and local features of nonstationary signals in the time domain and frequency domain [22]. In recent years, the random load spectrum processing technology, which integrates a variety of artificial intelligence (AI) technologies and modern signal processing methods, has been highly valued and extensively studied by scholars. Time–frequency analysis methods, such as short-time Fourier transform, wavelet transform (WT), and empirical mode decomposition (EMD), have become hot spots in the research of random load signal processing [23], [24], [25]. Some AI algorithms, such as machine learning (ML) and neural network (NN), have also been introduced into the field of random load processing and recognition. Solhmirzaei et al. [26] proposed a data-driven ML framework to effectively predict the failure mode of ultrahigh-performance concrete beams by utilizing ML algorithms, including support vector machine (SVM), artificial NNs (ANNs), and so on. Durodola et al. [27] used an ANN ML method to predict fatigue damage caused by random loads. Compared with the frequency-domain method, it has the advantages of fast analysis speed and good consistency with the rainflow counting method [27]. Therefore, we made full use of multidomain feature information and NN algorithms for road load recognition.

The main contributions of this article are listed as follows.

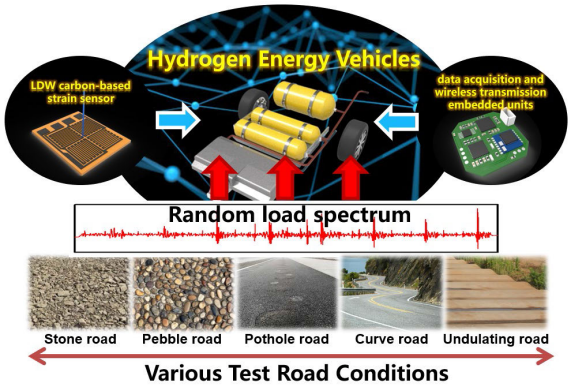

Fig. 1. Overview of the proposed work. Smart monitoring systems for HFCVs with innovative LDW carbon-based strain sensors, embedded data acquisition and transfer unit, and road surface pattern recognition ability.

1) This work demonstrated an LDW technology for fabricating flexible carbon-based sensors in a maskless, highly efficient, and scalable manner, with sensor performance superior to metal-based strain gauges.
2) The embedded low-power consumption data acquisition and wireless transmission unit developed in this research realized the signal acquisition, preprocessing, and wireless transmission of the LDW carbon-based strain gauges.
3) The proposed stacked pretrained deep NN (DNN) based on edge computing can disperse the pressure of system transmission and calculation and achieve high-precision monitoring and recognition of vehicle-mounted HSV loads under different road conditions.

The rest of this article is organized as follows. In Section II, we describe the fabrication procedure of the LDW strain sensor and its embedded data acquisition system integration. In Section III, we introduce the proposed data feature extraction and processing approach for road condition recognition. Section IV provides information on the installation of our measurement system and the random load bench test. The discussion of the experiments and the results can be found in Section V. Finally, conclusions are drawn in Section VI.

II. System Architecture

As a complex real-time monitoring technology, the SHM system architecture can be divided into the following subsystems: 1) sensor subsystem; 2) data acquisition, processing, and transmission subsystem; 3) data management subsystem; and 4) damage diagnosis and safety assessment subsystem [28]. In actual working conditions, most of the fatigue failures are closely related to the random load loading of road conditions. The various road conditions used for analysis are shown in Fig. 1. This work relied on the LDW carbon-based strain sensor with edge processing capabilities and the subsequent road condition load recognition algorithm to realize the SHM of HSV.

A. Sensor Fabrication Procedures

Fig. 2(a) is a schematic of a full-bridge strain sensor fabricated by LDW. The details of the preparation process and

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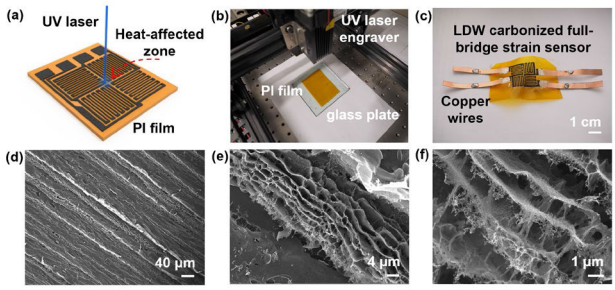

Fig. 2. Fabrication and characterization of LDW carbonized PI strain sensors. (a) Schematic of LDW fabrication process of a full-bridge strain sensor. (b) Photograph of LDW devices for sensor fabrication. (c) Photograph of LDW carbonized full-bridge strain sensor. (d)–(f) SEM images of the LDW-carbonized PI obtained at 3.15 W.

related results on the LDW strain sensors have been presented in the previous work [29]. The picture of the preparation process is shown in Fig. 2(b). The polyimide (PI) film with a thickness of 0.05 mm, supplied by Shenzhen Wangxing Adhesive Tape Company Ltd., was cleaned and firmly attached to a glass plate by adhesive tapes. Using a UV laser engraver (XG-1690A) with a wavelength of 405 nm as a heat source, the PI film was irradiated in an air environment to fabricate a carbon-based strain sensor without other pretreatment. The pattern of the strain sensor was designed by AutoCAD, and the carbonization and patterning processes were completed in one step through the 2-D programmed movement of the laser beam. Variations in laser power and scanning speed have impacts on the morphology, electrical, and mechanical properties of the carbon-based sensors. The laser power below 2.45 W would lead to partial carbonization of PI film, resulting in uneven material resistance, while the laser power above 3.5 W could cause severe distortion of the PI substrate due to thermal effect, resulting in excessive carbonization and embrittlement. The laser power and scanning speed were optimized to be 3.15 W and 7000 mm/min, respectively, for preparing a full-bridge strain gauge. The dimension of the full-bridge strain sensor fabricated by LDW carbonization method is 2 × 2 cm, as shown in Fig. 2(c). The carbonization of PI under laser irradiation is primarily accomplished through a photothermal mechanism [30]. The energy of the incident laser is absorbed by PI and converted into heat, generating extremely high temperature in the irradiated area, resulting in carbonization of PI. The scanning electron microscopy (SEM) images of the LDW carbonized PI are shown in Fig. 2(d)–(f). The grooves in Fig. 2(d) were generated by laser ablation of the PI film, with a 0.04-mm distance between adjacent LDW grooves. The cross sections of the groove [shown in Fig. 2(f)] present a porous stacked structure with fibrous structure connected between the stacked structures. During LDW process, PI was rapidly carbonized by laser irradiation, resulting in the instantaneous release of a large amount of pyrolysis gas, thereby forming a fluffy and porous structure [31]. The carbonization morphology of PI shown by SEM was consistent with the mechanism of laser carbonization of PI.

B. Hardware Design of the Embedded Unit

An embedded data acquisition and wireless transmission unit with edge computing ability was applied to replace

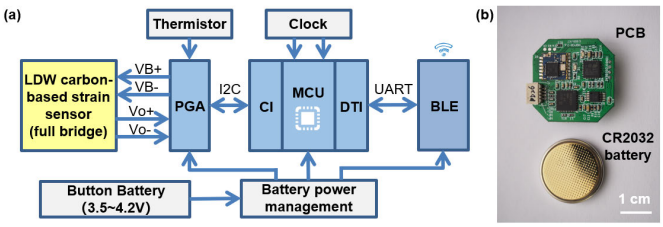

Fig. 3. (a) Circuit schematic of the embedded data acquisition and wireless transmission unit. (b) Photograph of the embedded unit.

conventional wired system for SHM. Fig. 3 shows the circuit schematic and corresponding printed circuit board (PCB) dimensions of the embedded data acquisition and wireless transmission unit. As shown in Fig. 3(a), the design of the data acquisition and transmission unit can be divided into four functional modules principally: signal acquisition module (signal conditioning and amplifier chip), signal processing module (micro processor unit (MPU) core controller), signal transmission module (Bluetooth wireless transmission), and energy supply management module. The signal acquisition module is a resistive sensor signal conditioner PGA305 with digital and analog outputs. The PGA305 device provides the interface and 2.5-V dc bridge excitation voltage for resistive bridge sensors. It has programmable analog front-end, analog to digital converter (ADC), and digital signal processing functions and can be directly connected to sensing devices. In addition, the PGA305 integrates a voltage regulator and oscillator to minimize the number of external components. The core processor adopts a 32-bit ultralow-power consumption microcontroller STM32L431 to process the voltage signal collected by PGA305 and manage the power module. The signal transmission module adopts a low-power Bluetooth dedicated wireless micro controller unit (MCU), CC2640R2F. It is mainly suitable for Bluetooth 4.2 and Bluetooth 5.0 low-power applications, with extremely low active RF and MCU current, ensuring excellent battery life. Since we attached importance to low power consumption performance in module selection and program code, the power module can achieve long-term power supply using batteries, such as CR2032 lithium manganese batteries.

In order to evaluate the low-power design of the embedded unit, we performed a calculation of the continuous working power consumption of the embedded unit (see detailed calculation process in supplemental material). The total power consumption of the embedded unit is 23.971 mW. Meanwhile, the power consumption of the embedded unit in the working state is measured by a digital multimeter DM3068 (RIGOL) and powered by a programmable dc power supply DP821A (RIGOL), as shown in Fig. S1. In the test, the embedded unit was connected to the full-bridge strain sensor, with all modules in working condition. The power supply voltage of the unit is 3.037 V, and the average current is 8.068 mA. The average power consumption obtained is 24.503 mW, which is close to the calculated result. Under the condition of continuous operation and regular collection of bridge voltage, a single unit could work continuously for about 30 h by using a 240-mAh CR2032 battery as power supply. The continuous operation of the PGA305, MCU, and Bluetooth modules are the major contributors to the power

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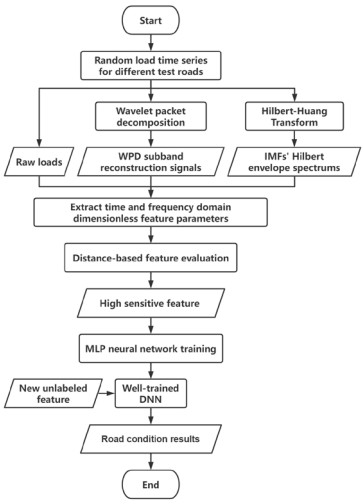

Fig. 4. Flowchart of random load pattern recognition method.

consumption of the unit. The transmission frequency and size of wireless data packets, the sampling period of sensors, and the calculation amount of the algorithm in processor play decisive roles in the power consumption of the device. In our future work, we will further reduce the power consumption of nodes by adding timer interrupts and optimizing algorithms. Data are preprocessed through distributed edge units and embedded algorithms, which will reduce the redundancy of raw data and retain useful data for transmission to base stations and centralized processing.

III. Methodology

The random load spectrum collected in actual vibration contains nonstationary, nonlinear, and complex modulation components, as well as invalid noise signals, which affects the accuracy of mechanical structure condition monitoring. Therefore, a DNN model was constructed, and two time–frequency-domain signal analysis methods were combined to process the random load spectrum.

The flow of the proposed random load signal processing is illustrated in Fig. 4. Three-layer wavelet packet decomposition (WPD) and Hilbert–Huang transform (HHT) were used to decompose the original load spectrum. A total of 150 time-domain and frequency-domain dimensionless feature parameters were extracted from the original load, WPD reconstruction signals, intrinsic mode functions (IMFs), and HHT envelope spectra. The sensitivity of the feature parameters was evaluated using a distance-based evaluation method. Subsequently, 20 features with the highest sensitivity were selected as the input features of the DNN to avoid invalid input features affecting the training speed, accuracy, and generality of the network. The network was trained on the random load spectra of six road conditions; then, the test samples of the six road conditions were input into the network for testing, which yielded the road condition recognition results. The proposed stacked pretrained DNN based on edge computing can disperse the pressure of system transmission and computing, effectively simplify the network structure, and can be subsequently deployed in embedded nodes relatively easily.

A. Feature Extraction

WPD is a technique to decompose a signal repeatedly into successive low- and high-frequency components. Specifically, the wavelet details of each layer are further decomposed into their respective approximation and detail components, which differs from wavelet analysis that only decomposes into the approximation components. Through the three-scale WPD, the original signal is decomposed into a series of interconnected and independent bands. The signal component of each frequency band subspace corresponds to the local time–frequency information of the original signal in the same frequency band, which can ensure the integrity of the signal. The reconstructed signal of each frequency bands shows the time-domain components of the original load signal in each frequency band.

HHT is an adaptive signal analysis technique, which is capable of processing nonlinear and nonstationary random signals. The random load signal is decomposed into multiple layers to obtain a series of stationary signal components with different scale characteristics, from which the load features contained in different IMFs are extracted. It consists of two parts: EMD and Hilbert transform. EMD can decompose a complex signal into a finite number of IMFs. The IMF component is an oscillating function with time-varying frequency, which can reflect the local characteristics of the nonstationary signals. Each decomposition extracts a high-frequency component of the signal and uses the remaining low-frequency component as a new signal. The signal is represented as N basic mode component and a residual. The Hilbert transform is then performed on each IMF and the Hilbert envelope spectrum is extracted. Finally, Fourier transform is performed on the Hilbert envelope spectrum of the IMFs to effectively extract the frequency-domain features of the signal.

Time-domain dimensionless feature parameters, such as skew factor, crest factor, impulse factor, margin factor, kurtosis factor, shape factor, and so on, can reduce the interference of the working environment and vehicle load and improve the accuracy and stability of load pattern recognition. Similarly, the extraction of dimensionless feature parameters in the frequency domain can more accurately describe the frequency-domain information of the load.

B. Feature Evaluation and Recognition

After obtaining the feature parameters in time domain and frequency domain, more feature parameters can usually express the signal more comprehensively. However, redundant feature parameters increase the complexity and training difficulty of the network model and even reduce the accuracy. Some feature parameters have higher sensitivity to classification and are independent of each other, which can be easily identified in fault classification. Feature subsets are extracted by introducing a distance-based evaluation method. The larger the average intercategory distance d_j^{outer} and the smaller average intracategory distance d_j^{inner} of a feature parameter, the more favorable it is for classification. Suppose that a J-dimensional feature vector set has C different categories, and the sample number of per category is Nc.

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The feature sensitivity evaluation steps are as follows:

1) Calculate the intracategory distance of the jth feature of the cth category through

$$d_{c,j} = \frac{1}{(N_c \times (N_c - 1))} \sum_{l,n=1}^{N_c} |q_{n,c,j} - q_{l,c,j}|$$
$$c = 1, 2, \ldots, C;\ j = 1, 2, \ldots, J$$
$$l, n = 1, 2, \ldots, N_c,\ \ l \neq n \tag{1}$$

where $q_{n,c,j}$ and $q_{l,c,j}$ represent the jth feature parameter of the nth and lth sample of the cth category, respectively. The equation of the average intracategory distance of the jth feature in C categories is obtained as

$$d_j^{(\text{inner})} = \frac{1}{C} \sum_{c=1}^{C} d_{c,j}. \tag{2}$$

2) Calculate the average of each feature as

$$u_{c,j} = \frac{1}{N_c} \sum_{n=1}^{N_c} q_{n,c,j}. \tag{3}$$

Then, the average intercategory distance of the jth feature in C categories can be calculated by

$$d_j^{(\text{outer})} = \frac{1}{C \times (C - 1)} \sum_{e,c=1}^{C} |u_{e,j} - u_{c,j}|$$
$$e, c = 1, 2, \ldots, C,\ \ e \neq c \tag{4}$$

where $u_{e,j}$ and $u_{c,j}$ represent the average of the jth feature of the eth and cth category, respectively.

3) Finally, the distance evaluation factor of the jth feature was determined using the following equation:

$$\alpha_j = \frac{d_j^{(\text{outer})}}{d_j^{(\text{inner})}}. \tag{5}$$

Factor α reflects the measure of sensitivity of each feature parameter. A larger α_j indicates that the jth feature parameter is more sensitive to classification and can be set as a sensitive feature.

Finally, a multilayer perceptron (MLP) DNN model was built to classify and identify the sensitive feature parameters of each road conditions extracted in the above process. Shallow perceptron models encounter difficulties in solving linear inseparable problems, so multiple shallow perceptron need to be stacked to form an MLP. Generally, an MLP network comprises an input layer, an output layer, and multiple hidden layers, with each layer consists of multiple neurons. The difference from the traditional shallow learning model is that: 1) the depth of the model structure increases (usually more than five layers) and 2) the importance of feature learning is emphasized. The feature representation of the sample in the original space is transformed into a new feature space through layer-by-layer feature transformation, making the classification easier. Therefore, a DNN multilayer structure with one input layer, six hidden layers, and one output layer was constructed. The number of hidden neuron nodes in each hidden layer was 20. The rectified linear unit (ReLU) function was selected as the nonlinear activation function, which could better overcome the problem of gradient disappearance. As road condition recognition was a multiclassification problem, the Softmax function was utilized as the classification function to calculate the probability of each target category among all possible target categories. The Adam method was selected as the network training method and the categorical-crossentropy function was utilized as the loss function.

Based on the methodology, the road random load spectrum is analyzed, and the results are presented in Section V.

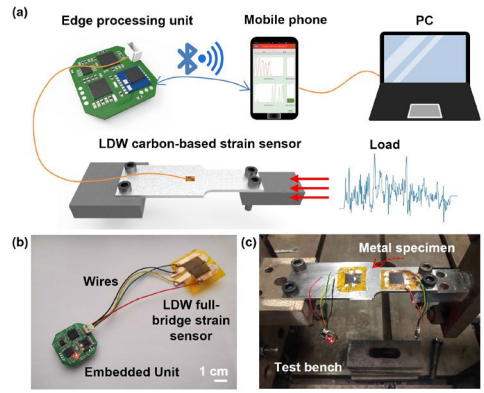

Fig. 5. Experimental setups. (a) System architecture. (b) Sensor and embedded unit subsystem. (c) Bench test.

IV. Experimental Setup

The entire monitoring system under random load experiment [shown in Fig. 5(a)] was composed of LDW carbon-based strain sensors, edge embedded units, mobile phones, and a PC. Fig. 5(b) shows the picture of the LDW sensor and the embedded unit. Furthermore, an android application (APP) was developed to control the embedded monitoring system of HSVs. The APP connected to the Bluetooth module of the edge monitoring unit through the Bluetooth of the mobile phone and supported functions, such as real-time stress data display, historical stress data query, stress data storage, and edge unit battery power monitoring.

Fig. 5(c) shows the picture of the monitoring system attached to the metal specimens during a random load bench test. The hydraulic servo fatigue testing machine (Mechanical Testing and Simulation Ltd.) was used as the random load test device. Bench testing machine is mainly composed of loading system, control system, measuring system, and other parts. In this experiment, random load spectra of six road conditions were applied to the metal specimens by the bench test machine. These load signals were obtained through real vehicle tests under six road conditions of stone road, pebble road, undulating road, pothole road, W-shaped curve, and P-shaped curve in the motor vehicles testing ground. The loading times of random loads in different road conditions are different. The stone road, pebble road, and pothole road are all loaded 200 times, and the P-shaped curve, undulating road, and W-shaped curve are loaded 170, 100, and 30 times, respectively. The strain range of the load data under each test road condition is generally 0.005%–0.1%, and the frequency range is 0–85 Hz. However, due to the existence of tires and suspension structures, high-frequency vibrations suffered

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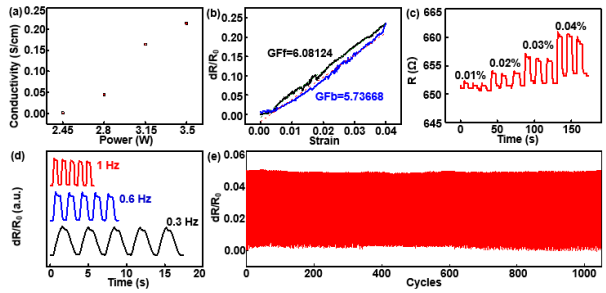

Fig. 6. Electrical performance of the LDW carbon-based strain sensor. (a) Conductivity of the sensor at different laser powers. (b) Relative resistance curve of the sensor for applied strain. (c) Response curves of the minimum detectable strain of the sensor. (d) Response curves of the sensor under different frequencies of strain application. (e) Mechanical durability test of the sensor at $\varepsilon \approx 1\%$.

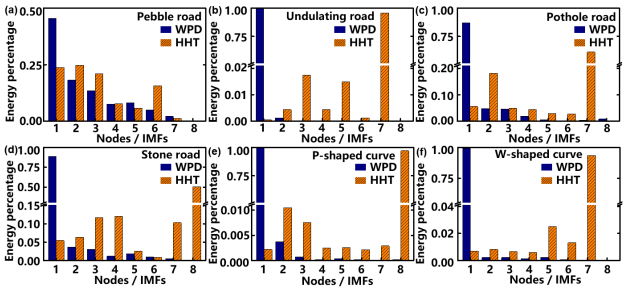

Fig. 7. Energy feature distribution of WPD reconstructed time-domain signals and IMF in frequency domain under random load spectrum of six road conditions. (a) Pebble road. (b) Undulating road. (c) Pothole road. (d) Stone road. (e) P-shaped curve road. (f) W-shaped curve road.

by the body and cylinders are greatly reduced and can even be ignored in many cases [32]. Thus, the impact of low-frequency (1∼3 Hz) loads on the body and gas cylinders was considered in this study. The bench test fixtures for the steel plate were installed in accordance with the tension and compression test method. The LDW full-bridge strain sensor was then attached to the stress concentration position of the steel plate. The power rod applied random tension and compression loads to the steel plate. The temperature was kept constant at 25 °C throughout the bench test. When acquiring the random load on the steel plate, the edge unit performed filtering and feature extraction on the random load spectrum and wirelessly transmitted the collected data to the mobile phone via Bluetooth for real-time display and data storage. Finally, feature evaluation and NN training were conducted on the extracted feature parameters using a personal computer.

V. Experimental Results and Discussion

A. Electromechanical Characteristics of the Carbon-Based Strain Sensor

In order to evaluate the sensing performance, the conductivity of the porous graphite generated by the LDW was first explored. Four-point probe instrument (ST2253, Suzhou Jingge Electronic Company Ltd.) was used to obtain the conductivity of the LDW generated carbon. As shown in Fig. 6(a), the conductivity gradually increases from 0.00153 to 0.21394 S/cm with the increase of laser power. Using the selected fabrication parameters in Section II, the conductivity of the LDW carbon-based strain sensor is 0.16418 S/cm. The sensing performance of the LDW carbon-based strain sensor was characterized with an electrochemical workstation (CHI660E, Shanghai Chenhua Instruments). The applied strain was loaded by a Mark-10 force meter on a motorized test stand (ESM303, Mark-10 Inc.). The corresponding electromechanical response of the device is shown in Fig. 6(b). The electrical curve of the sensor was linear over a range of 4% strain. The sensor owned a sensitivity of 6.08 at $\varepsilon < 4\%$, which was higher than that of traditional metal strain gauges (∼2). When deformation was applied to the LDW carbon-based strain sensor with porous stack structure, the loss of the conduction pathway significantly affected the resistance of the sensor, so a high gauge factor could be obtained. The minimum strain detection limit of the sensor is shown in Fig. 6(c). The sensor could respond to a minimum strain change of 0.01%, indicating that it could be used for small deformations detection of mechanical structures. In addition, the strain sensor could respond to the strain with a frequency up to 1 Hz [shown in Fig. 6(d)] and could detect the load state change of the mechanical structure in time. A durability test on the LDW carbon-based sensor was also conducted by performing 1000 cycles of tensile tests under a load of 1% strain. Fig. 6(e) shows that the resistance change is small after 1000 cycles of testing, indicating a good stability of the device.

B. Performance Measure of WPD and HHT Envelope Spectrum

Fig. 7(a)–(f) shows the energy distribution of the random load spectra under six road conditions obtained by WPD and HHT, respectively. Through the ten-order Daubechies discrete orthogonal wavelet db10, the original load spectrum was decomposed and reconstructed by three-scale WPD. As a result, reconstructed time-domain signals in eight frequency bands were obtained, and each contained the time-domain components of the original load spectrum in the corresponding frequency band. The energy of the random load spectrum was mainly concentrated in the low-frequency band, indicating that the random load during vehicle driving was primarily composed of low-frequency signals. The frequency band energy distribution of the load spectrum was noted to be different under varying road conditions. For example, the energy in higher frequency band of WPD under the pebble road condition [Fig. 7(a)] is higher compared with other road conditions, which shows that the WPD signals contain characteristic information of different loads. Comparing with the time-domain energy feature distribution of WPD, the frequency-domain energy feature distribution of the HHT envelope spectrum can differentiate the six random load spectra more effectively. HHT is helpful to extract the feature parameters of the random load signal and can effectively establish the mapping relationship between the road load pattern and the feature parameters in the frequency domain.

C. Analysis of the Feature Selection

After performing WPD and HHT processing on the original load signal, feature sensitivity evaluation was conducted on the

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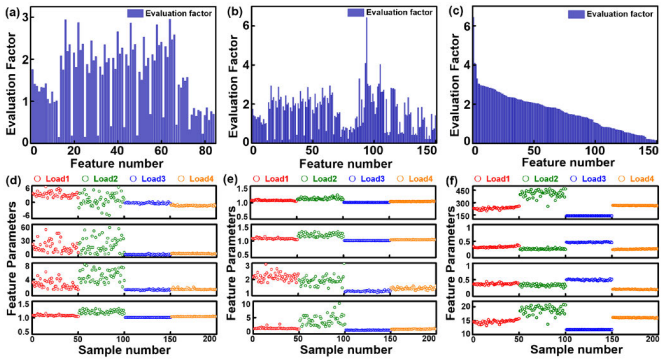

Fig. 8. Evaluation factors of (a) 84 time-domain nondimensional feature parameters, (b) 150 time–frequency-domain feature parameters, and (c) 150 feature parameters sorted from high to low. The comparison of the classification effect of the four dimensionless feature parameters extracted from (d) six time-domain feature parameters, (e) 84 time-domain feature parameters, and (f) 150 time–frequency-domain feature parameters under four types of road surfaces loads.

extracted time-domain and frequency-domain feature parameters. Fig. 8(a) shows the evaluation factors of 84 time-domain feature parameters extracted from the original loads, WPD signals, and EMD signals; Fig. 8(b) demonstrates the evaluation factors of 150 time–frequency-domain feature parameters obtained from the original loads, WPD signals, EMD signals, and HHT envelope spectra. The evaluation factors were arranged in descending order, as shown in Fig. 8(c). The evaluation results indicated that some dimensionless feature parameters in the frequency domain obtained from the original signal and HHT envelope spectrum had higher sensitivity than those in the time domain.

Fig. 8(d)–(f), respectively, demonstrated the distinguishing capacity of the feature parameters extracted by three random load processing methods on four road conditions, illustrating the effect of feature sensitivity evaluation. In detail, each processing method extracted four feature parameters, and the samples of each feature parameter were composed of four road conditions, with 50 samples for each road condition, resulting in a total of 200 samples. The four road conditions were pebble road, pothole road, W-shaped curve, and P-shaped curve, which were recorded as Loads 1–4, respectively. As shown in Fig. 8(d), four time-domain dimensionless feature parameters of skewness, kurtosis, crest factor, and shape factor were extracted from the original load data. The feature parameters extracted from 50 samples of the same road condition in Load 1 and Load 2 had poor consistency. The ability of the four feature parameters to discriminate four road condition loads was also poor, especially in Load 3 and Load 4. Fig. 8(e) showed the four feature parameters with the highest sensitivity among the 84 time-domain dimensionless features extracted from the original load data, wavelet packet band components, and IMFs. The consistency of the feature parameters under the same road condition in Load 1 and Load 2 had been improved, yet the discrimination ability for Load 3 and Load 4 was barely improved. In addition to the 84 time-domain features, 66 frequency-domain feature parameters were extracted from the original load and HHT envelope spectra. The four highest sensitivity feature parameters among

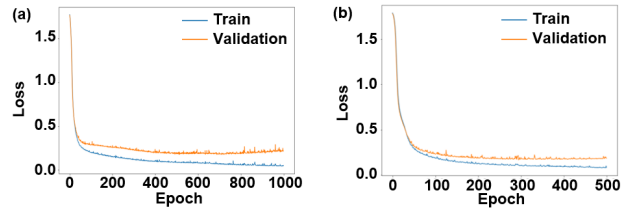

Fig. 9. Trending curve of training set and validation set loss for MLP at (a) 1000 epochs and (b) 500 epochs.

the 150 time–frequency-domain dimensionless features are shown in Fig. 8(f). The consistency of the feature parameters in Load 1 and Load 2 was improved, and the discrimination for all four loads was also significantly improved. It showed that the extracted frequency-domain feature parameters were more consistent under the same road condition and more sensitive to different load classifications, which was in line with the feature sensitivity evaluation results. The feature evaluation could evaluate the effectiveness of the time- and frequency-domain analysis methods, and the processing method with high feature sensitivity can be applied in the edge processing programs to simplify the processing flow. For example, it is not necessary to extract time-domain features in the first frequency band signal of WPD; and only the first three IMFs need to be obtained for EMD and HHT envelope spectrum.

D. Performance of the Random Load Pattern Recognition

In this section, we evaluated the performance of the proposed road condition load spectrum recognition method based on sensitive features and DNN network and compared it with some SOTA methods.

A feature parameter dataset was first established. In the bench test, a total of 900 random load spectra of six road conditions were acquired using the LDW carbon-based strain sensor. The random load pattern recognition model utilized the time–frequency-domain feature parameters of the random load as the input and the test road condition types as the output. After evaluating the sensitivity of 150 time–frequency-domain feature parameters according to the distance-based evaluation method, the top 20 features with highest sensitivity were selected as the input feature vector of the NN for training. The corresponding 900 feature parameter datasets were extracted from 900 load spectrum samples and allocated into training set and test set at a ratio of 3:1. The training set data were further divided into training data and validation data at a ratio of 3:1. Based on the *Tensorflow* platform, an MLP with a 20-D input and a 6-D output was constructed and was trained using the training and validation sets. As shown in Fig. 9(a), the loss of the network on the training and validation sets changes during training process. The total loss of the training set dropped from 1.775 to 0.049 after training to 1000 epochs. The total loss of the validation set decreased initially, reached a plateau at around 500 epochs and returned to around 0.215 after training up to 1000 epochs, indicating a slight overfitting phenomenon. Consequently, 500 epochs were selected as the optimal network parameters, and the variation trend of loss with the training epochs is shown in Fig. 9(b).

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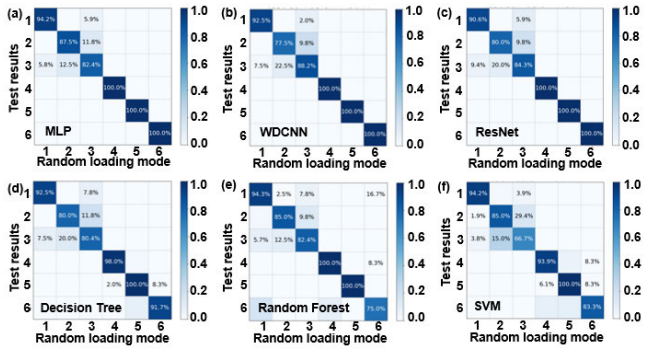

Fig. 10. Confusion matrix of all the models. (a) Proposed MLP network. (b) WDCNN. (c) ResNet. (d) Decision tree. (e) Random forest. (f) SVM.

TABLE I

COMPARISON OF THE SOTA METHODS USED FOR ROAD CONDITION RECOGNITION

Method	Accuracy (%)
WDCNN	91.67
ResNet	90.79
Decision Tree	89.47
Random Forest	89.91
SVM	86.40
Proposed work	92.5

We use some SOTA methods to process the load spectrum of six road conditions, and the performance of recognition is shown in Fig. 10. The random load recognition results of the proposed MLP network on the test set are shown in Fig. 10(a), and the recognition accuracy reaches 0.925. After comparing this method with some SOTA methods (as shown in Table I), we found that our recognition accuracy has reached or exceeded some road anomaly detection and classification methods based on feature extraction technology and deep learning technology. This demonstrates the feasibility and effectiveness of the proposed random load pattern recognition method on test roads.

VI. CONCLUSION

In conclusion, we developed an edge intelligent monitoring system for monitoring the load stress state of HSVs in driving HFCVs based on LDW carbon-based strain sensors, embedded systems, and AI algorithms. The LDW carbon-based strain sensor had high sensitivity (6.08) and high response speed (1 Hz), effectively measuring the strain status under varying road conditions. A miniaturized, highly integrated data acquisition and wireless transmission embedded unit was developed. Experimental tests verified the feasibility and integrality of the monitoring system, which also proved the effectiveness of the fusion time–frequency-domain signal analysis method in extracting the features hidden in signal. The distance-based feature evaluation method could effectively screen the feature parameters, optimize the structure of MLP, and improve the accuracy of recognizing random load types (up to 92.5%). We believe that this research can effectively promote the design of SHM system for HFCVs.

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